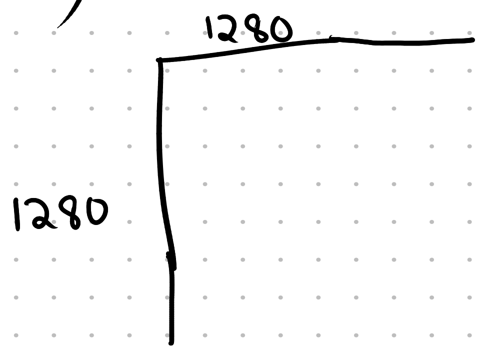
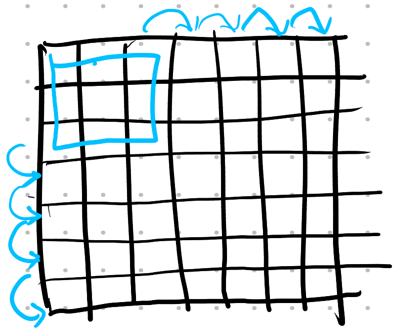
Q4) To downscale an img: we will use mean filter





5×5

$= 25$ is out

$7 \times 7 = 49 \longrightarrow 25$

$N^2 = 49 \longrightarrow (N-2)^2$

filter size $= f$

$N^2 = 1280 \Rightarrow N = 35.77$

$(N-f)^2 = 300$

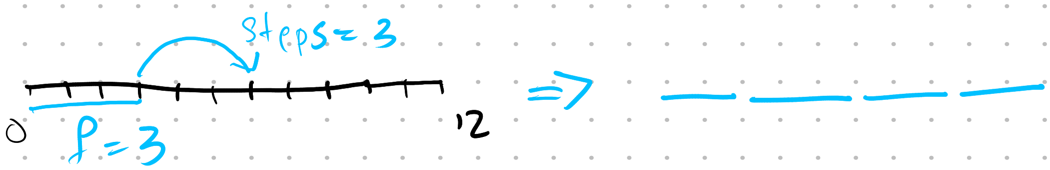
$(35.77 - f)^2 = 300 \;\big|\; 35.77 - f = \sqrt{300} = 17.32$

or $f = 35.77 - 17.32 = \underline{\underline{18.45}}$

But then it's just skipping the initial part.

So we got to adjust in steps

It's better if no. of steps = filter width



$12 \times 12 \longrightarrow 4 \times 4$

$144 \longrightarrow 16$

So point is no. steps = filter size

So $f \times f = 300 \Rightarrow f = \sqrt{300} \approx 17.32$ for a 300×300

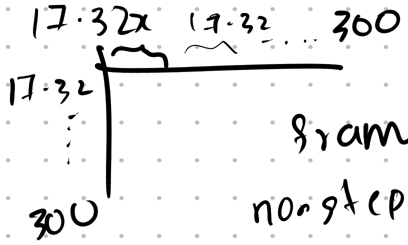
pixels: $90{,}000 \rightarrow 289$

So $1280^2 = 16,38,400 \longrightarrow 90,000$

So we want to scale down to 18.204 times

What is f2 for 1280x1280 s.t output = 300x300

i.e. $f_{300} \times 18.204$



frame = 17.32

non step = 17.32

output = 17.32^2

1) Do padding of 20 on all sides.
2) filter with 300 steps & frame ≈ 300

$$300 = \frac{1280}{x}$$

$x = f + ns$ — n times x step size, f → filter len, $n \sim f$

$\Rightarrow 4.276 = n + ns$

$= n(s+1)$

or $\frac{4.276}{n} = s+1$

If img = 1200

the $x = 4^2$

i.e. $n + n^2 = 16$

i.e. $n(1+n) = 16$

$\Rightarrow 3 \times 4 = 16$

i.e. $f = 3 \times 3$ & stepsize = 4

or $s = \frac{4.276}{n} - 1$ — If $s \sim n$

$n^2 = 4.276 - 1$

or $n = \sqrt{3.276} = \underline{\underline{1.810}}$

$3 + 299 \times 4 = 1199$

↓ f, ↓ step, ↓ stepsize

− 20 for padding

$\underline{\underline{20 \text{ filter size}}} + 300 \times 4$

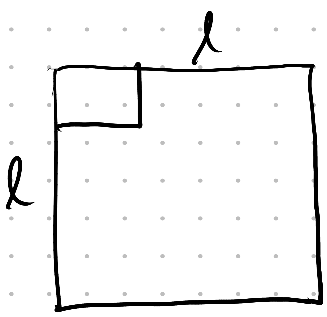
↓ step size

1) Create 20 x 20 mean filter

2) Apply the filter with stepsize 4 & 300 steps in x & y

Logic of applying filter

output = []




```
K = len(mean_filter)
i = 0
while i+k <= l:
    row_out = []
    j = 0
    while j+k <= W:
        s = 0
        for ki in range(k):
            for kj in range(k):
                s +=
```

$$\text{Output ht} = \left\lfloor \frac{H-k}{\text{stride}} \right\rfloor + 1$$

$$\text{Output wt} = \left\lfloor \frac{W-k}{\text{stide}} \right\rfloor + 1$$